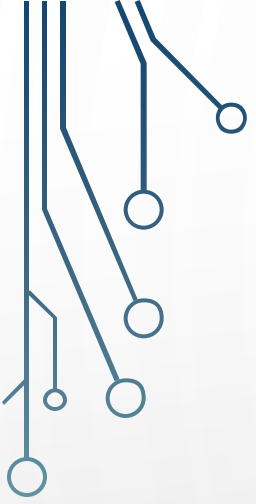


A decorative graphic on the left side of the slide consists of a network of thin, light blue lines. These lines form a complex, branching pattern that resembles a circuit board or a neural network. The lines are of varying thickness and connect to small, hollow circles at various points, creating a sense of connectivity and flow. The overall style is clean and modern, with a focus on geometric shapes and a limited color palette.

ELECTRONIC PRODUCTS ANALYSIS REPORT

PRESENTED BY LINDA WACHIRA



DATASET OVERVIEW

About this dataset

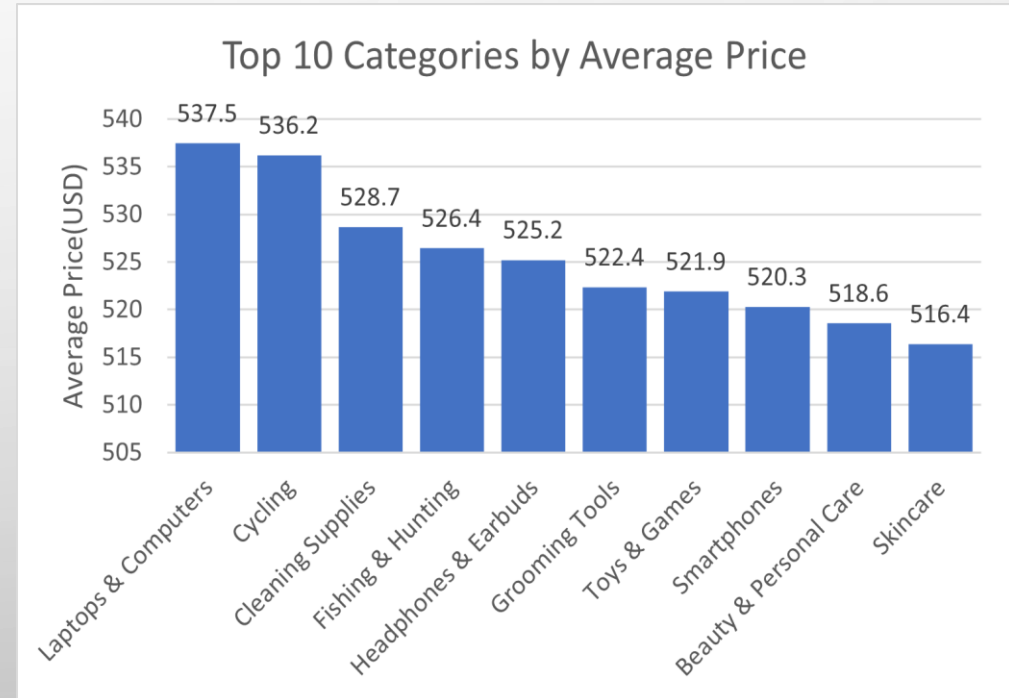
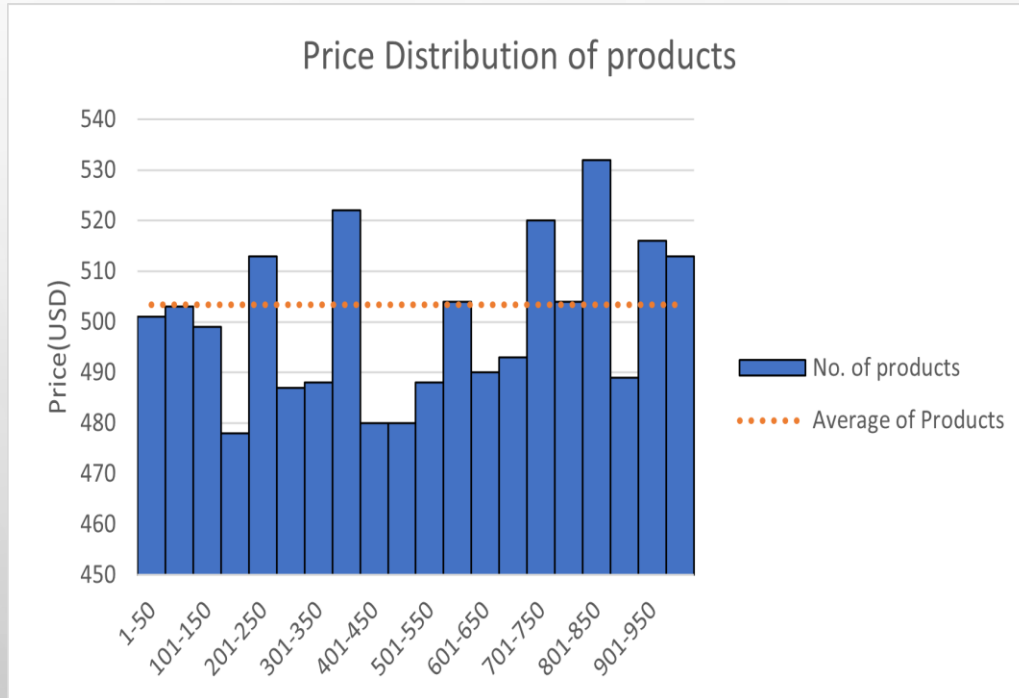
This is an e-commerce electronics and lifestyle products catalogue with **10,000 rows** and **13 columns**. It covers **34 categories**, including tech gadgets like **smartwatches and laptops**, **fitness equipment**, and **clothing**. Each record includes a product name, brand, price, stock amount and availability, color, size, and unique IDs.

Three questions guided the analysis of this dataset:

1. What is the price distribution of products?
 2. What is the status of the stock inventory?
 3. Which categories represent the greatest total value of inventory?
- 
- 
- 

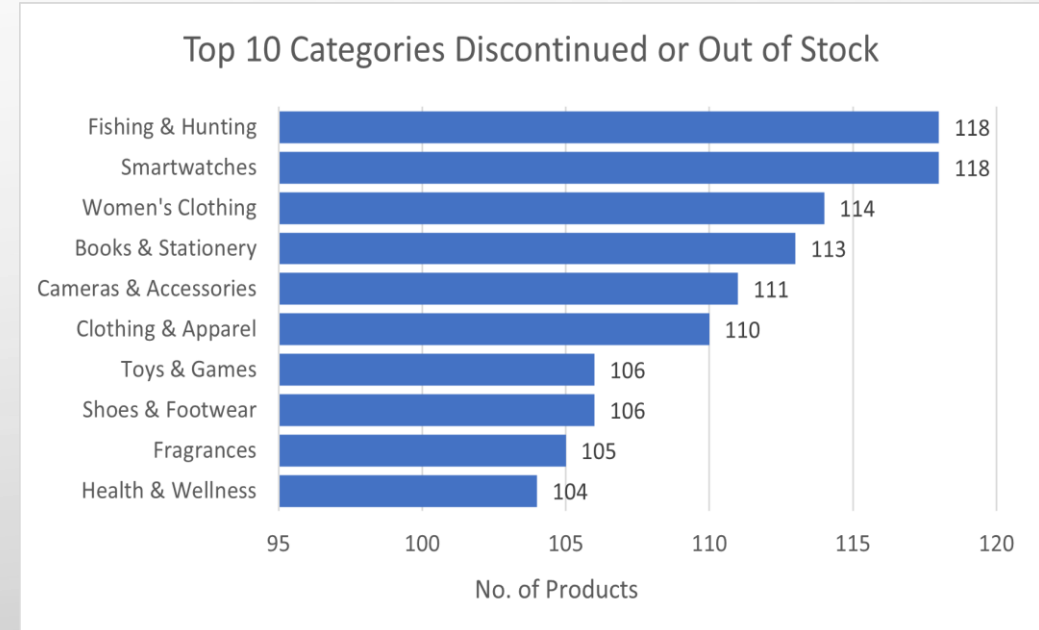
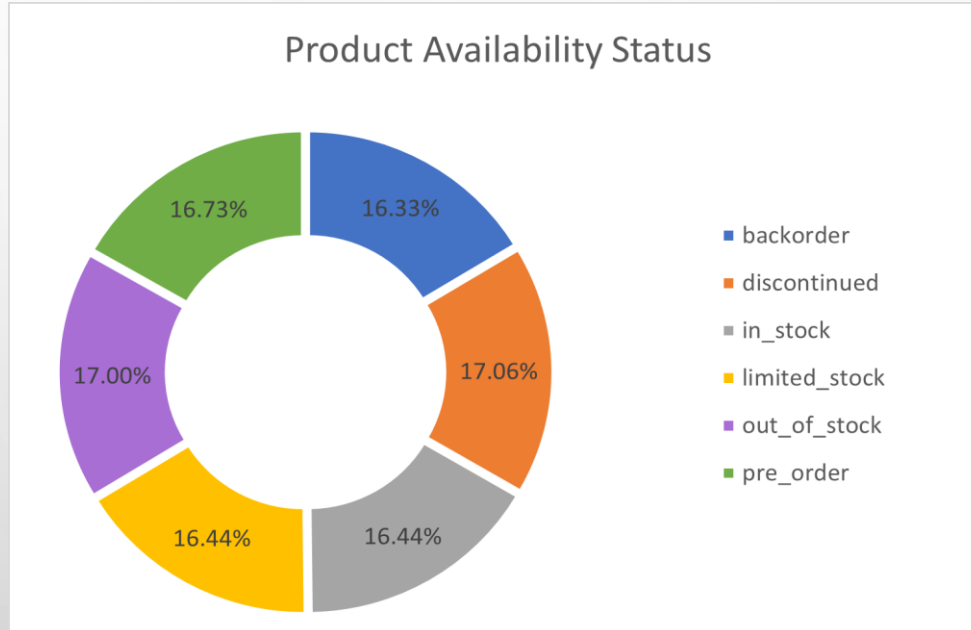
KEY FINDINGS

Q1: What is the price distribution of products by category?



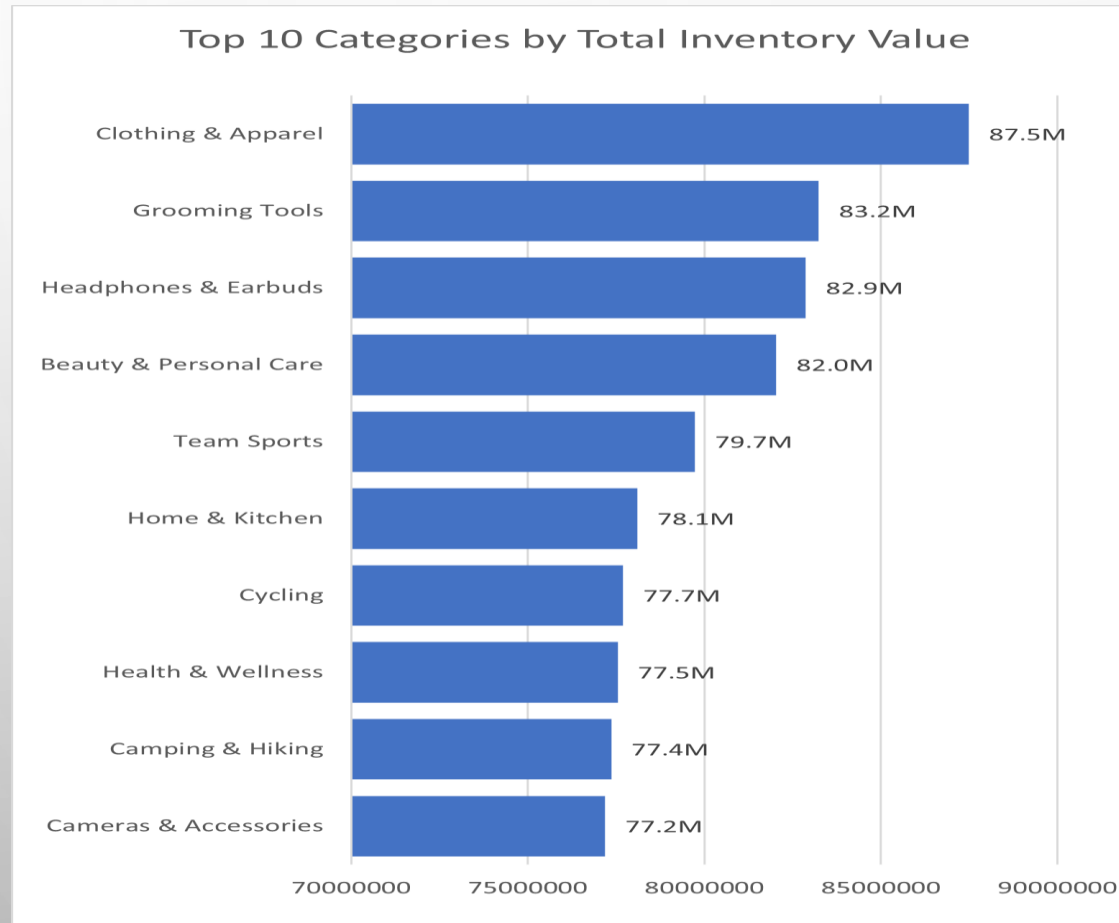
- Prices are uniformly spread from \$1 to \$999.
- The mean (\$503) and median (\$506) indicate a uniform distribution.
- Laptops & Computers take the lead with \$537, but not in the dramatic sense, as from the highest to the least(\$537-\$475) the spread is uniform by only \$62..

Q2: What is the status of the stock inventory?



- Availability is almost equally split across all 6 statuses.
- 34% of the stock inventory is either out of stock or discontinued.
- Fishing & hunting and Smartwatches lead with 118 products each.

Q3: Which categories represent the greatest total value of inventory?



- Clothing & Apparel leads with \$87.5M worth of stock.
- Majority of the top 10 categories are lifestyle-based.
- Combined inventory value of top 10 sums up to \$803M.
- The average total value across all 34 categories is \$25M.

INSIGHTS, LIMITATIONS & ASSUMPTIONS

Insights

- The catalog is broad(34 categories ranging from 260-320 products each) with uniform prices spanning \$1-\$999.
- 34.1% of the product listings are unavailable, a significant gap that requires attention, with SmartWatches and Fishing & Hunting being the most affected(118 each).
- Clothing & Apparel leads the inventory total value with \$87.5M, outdoing all tech categories despite having lower average prices.

Limitations & Assumptions

- The uniform distributions are unrealistic and don't reflect real market behavior.
- Descriptions hold no context; therefore, further analysis, like sentiment analysis, cannot be done.
- Absence of date fields makes actions of trend analysis or time-series forecasting impossible.
- Brand analysis cannot be done with 9241 brands out of 10,000 products, as a majority of these brands appear only once.
- The internal ID is unreliable as they are only 99 across 10,000 rows, making the EAN the only reliable unique ID.